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User Guide for the DEN Paper Template

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Abstract

An abstract summarizes, in one paragraph of 300 words or fewer, the major aspects of the entire paper. It should include: 1) the overall purpose of the study and the research problem you investigated; 2) the basic design of your research approach; 3) major findings as a result of your analysis; and 4) a brief summary of your interpretations and conclusions.

Keywords: keyword; keyword-two; keyword number three

JEL Classification: C22, O47

1. Pendahuluan

2. Studi Literatur

Trade liberalization and welfare has been the subject of a number of empirical research. Standard trade models predict aggregate welfare gains from the removal of trade barriers, where liberalization allows economies to specialise according to comparative advantage and open consumers to access of more varied goods (Krugman, Obstfeld and Melitz, 2022). These aggregate predictions, however, say little about how the gains are distributed, and the empirical literature has increasingly turned to that question.

Much of the recent evidence shows that the direction of the effect depends on which tariffs are cut. Kis-Katos and Sparrow (2015) combine the annual manufacturing survey (Survei Industri or IBS) with Sakernas for the period 1993 to 2002 and estimate effects at both the provincial and the district level in Indonesia. Both levels of analysis indicate that average wages rose in response to reductions in tariffs on intermediate inputs. The two tariff types also move poverty in opposite directions, since lower output tariffs are associated with higher poverty across the poverty measures while lower input tariffs are associated with poverty reduction.

Amiti and Konings (2007) find that a 10 percentage point fall in input tariffs raises productivity by about 12 percent for firms that import their inputs, at least twice the gain from an equivalent cut in output tariffs. To be precise, importers receive this full benefit while non-importers receive only about 3 percent, so the gains from input liberalization are concentrated among a subset of firms rather than shared evenly across the sector.

Evidence from Brazil points to a further source of heterogeneity, this time across labour market segments rather than across firms. Paz (2014) finds that the gains accrued disproportionately to formal employment. A one percentage point reduction in the tariffs applied by trading partners to Brazilian exports raised formal-sector wages by roughly 0.32 percent, whereas a one percentage point reduction in Brazil's own import tariffs lowered formal wages by about 0.05 percent. Lower partner tariffs also reduced the probability that a worker was employed informally.

The evidence is not uniformly positive. Delesalle et al. (2025) find no significant average effect of the ASEAN-China Free Trade Area on the probability of holding a wage job in Indonesia. The estimated overall effect is economically very small or statistically insignificant for men, for women, and for less-educated workers. It is larger and statistically significant only for educated workers, for whom the average increase is 1.9 percentage points against a baseline probability of 17.4 percent in 2000. Since this group accounts for around a quarter of the workforce, the distributional incidence of the agreement was regressive. Topalova (2007) reports a sharper result for India, where a district exposed to the average tariff reduction recorded a poverty headcount roughly two percentage points higher and a poverty gap roughly 0.6 percentage points higher than a district whose tariffs were unchanged.

Whether such adverse effects materialise appears to depend on the institutional setting. Goldberg and Pavcnik (2003) find that Colombian tariff reductions raised informal employment by roughly 0.1 percentage points per percentage point of tariff cut, but only before Law 50 of 1990 reduced firing costs. Post-reform, the interaction between tariffs and informal employment is consistently positive, indicating that the same tariff decline leads to a smaller increase in informality afterwards, and possibly a decrease. These results indicate that trade policy moves informality only when formal employment is costly to adjust.

Goldberg and Pavcnik (2007) find that globalization increases the wage skill premium in developing countries at the economy-wide level, within industries, and within firms. Amiti and Cameron (2012), however, report the opposite for Indonesian manufacturing. Using firm-level data with firm and year by island fixed effects, they find that a 10 percentage point reduction in input tariffs lowered the within-firm skill premium by about 1.4 to 2.6 percent, while output tariffs had no significant effect once input tariffs were controlled for. The effect is concentrated among firms that actually import intermediates.

In analysing the effects of trade liberalization on poverty and labour market outcomes, Kis-Katos and Sparrow (2015) construct district-level tariff exposure by combining nationally determined sectoral import tariffs with each district's initial economic structure prior to trade liberalization. This approach follows a **shift-share (Bartik) framework**, in which national tariff changes are weighted by predetermined district-level employment or industrial output shares.

For the poverty analysis, district output tariff exposure is measured using the initial employment distribution across sectors. Specifically, the tariff exposure of district k in year t is defined as

$$Tariff_{kt} = \sum_{s=1}^{S} \left(\frac{Q_{sk,1990}}{Q_{k,1990}} \times Tariff_{st} \right),$$

where:

- $Q_{sk,1990}$ denotes employment in output sector s in district k in 1990.
- $Q_{k,1990}$ is the total labour force in district k in 1990.
- $Tariff_{st}$ is the national import tariff applied to sector s in year t.
- S denotes the total number of tradable sectors.

For the firm-level analysis, Kis-Katos and Sparrow (2015) alternatively construct an output tariff measure using the initial industrial structure of each district. In this case, sectoral tariffs are weighted by each sector's share in the district's total industrial output in 1993, $\left(\frac{Q_{sk,1993}}{Q_{k,1993}}\right)$, rather than by employment shares.

To capture the effects of tariffs on intermediate inputs, the authors also construct an input tariff measure. The input tariff for each industry is first calculated as the weighted average tariff on all imported intermediate inputs, where the weights are given by each input's share in the industry's total input use. These industry-specific input tariffs are then aggregated to the district level using the district's initial employment shares:

$$InputTariff_{kt} = \sum_{s=1}^{S} \left(\frac{Q_{sk,1990}}{Q_{k,1990}} \times \sum_{j=1}^{J} \left(\frac{M_{js,1990}}{M_{s,1990}} \times Tariff_{jt} \right) \right).$$

where:

- $M_{js,1990}$ denotes the value of intermediate input j used by industry s in 1990.
- $M_{s,1990}$ is the total value of intermediate inputs used by industry s in 1990.
- $\frac{M_{js,1990}}{M_{s,1990}}$ represents the share of input j in the total intermediate inputs of industry s.
- $Tariff_{jt}$ denotes the national tariff on intermediate input j in year t.
- J represents the total number of input sectors.

Consistent with Kovak (2013), Kis-Katos and Sparrow (2015) exclude non-tradable sectors from the construction of the output tariff measure because these sectors are not directly exposed to international trade.

The effects of trade liberalization on poverty and labour market outcomes are then estimated using the following first-difference specification:

$$\Delta y_{kt} = \alpha + \beta_1 \Delta OutputTariff_{kt} + \beta_2 \Delta InputTariff_{kt} + \Delta X'_{kt}\gamma + I'_k\theta + \lambda_{rt} + \Delta\varepsilon_{kt},$$

where:

- Δy_{kt} denotes the change in the district-level outcome variable, including measures of poverty, employment, and wages.
- $\Delta OutputTariff_{kt}$ represents the change in district-level output tariff exposure.
- $\Delta InputTariff_{kt}$ represents the change in district-level input tariff exposure.
- ΔX_{kt} is a vector of time-varying control variables, including the share of the rural population, the share of the working-age population (16–60 years), adult literacy rates, and minimum wages.
- I_k is a vector of initial district characteristics, including the 1990 labour shares used as tariff weights, aggregated to one-digit sectors, the 1990 rural population share, and, in some specifications, the initial level of the dependent variable.
- λ_{rt} denotes island-by-year fixed effects.
- $\Delta\varepsilon_{kt}$ is the error term.

The first-difference specification removes time-invariant district characteristics and exploits within-district variation in tariff exposure over time. By combining predetermined employment shares with nationally determined tariff changes, the Bartik framework identifies heterogeneous district exposure to trade liberalization while reducing concerns regarding reverse causality.

Unlike Kis-Katos and Sparrow (2015), who estimate the model using a first-difference OLS specification, Delesalle et al. (2025) employ a Bartik instrumental variable (IV)strategy to address the potential endogeneity of tariff reductions. District-level exposure to trade liberalization depends on both the pre-liberalization industrial structure and industry-level tariff changes. Rather than examining multilateral trade liberalization, the authors focus exclusively on the bilateral tariff reductions between Indonesia and China under the ASEAN–China Free Trade Agreement (ACFTA), using pre-ACFTA industry shares to construct the exposure measures. Specifically, they consider three channels through which trade liberalization may affect local labor markets: output import tariff cuts, input import tariff cuts, and export tariff cuts.

The output import tariff measure is constructed as

$$\Delta OutputImportTariff_{d,2000-2010} = \sum_{s} \frac{L_{s,d,2000}}{L_{d,2000}} \cdot \Delta Tariff^{Indo}_{s,2000-2010}$$

where:

- $L_{s,d,2000}$ is the number of working-age individuals employed in industry s in district d in 2000.
- $L_{d,2000}$ is the total working-age population employed in district d in 2000.
- $\Delta Tariff^{Indo}_{s,2000-2010}$ denotes the change in Indonesia's import tariff on Chinese products in industry s between 2000 and 2010.

To capture the effect of imported intermediate inputs, the authors construct the input import tariff measure:

$$\Delta InputImportTariff_{d,2000-2010} = \sum_{s} \frac{L_{s,d,2000}}{L_{d,2000}} \left(\sum_{j} \frac{M_{j,s,2000}}{M_{s,2000}} \cdot \Delta Tariff^{Indo}_{j,2000-2010} \right)$$

where:

- $M_{j,s,2000}$ denotes the value of intermediate input j used by industry s.
- $M_{s,2000}$ is the total intermediate inputs used by industry s.
- $\frac{M_{j,s,2000}}{M_{s,2000}}$ represents the cost share of input j in industry s.
- The inner summation calculates the industry-level input tariff, while the outer summation aggregates these tariffs to the district level using initial employment shares.

The export tariff measure follows the same specification as the output import tariff but replaces Indonesian import tariffs with Chinese tariffs imposed on Indonesian exports:

$$\Delta ExportTariff_{d,2000-2010} = \sum_{s} \frac{L_{s,d,2000}}{L_{d,2000}} \cdot \Delta Tariff^{China}_{s,2000-2010}$$

where $\Delta Tariff^{China}_{s,2000-2010}$ denotes the change in Chinese import tariffs on Indonesian exports in industry s.

The relationship between tariff exposure and labor market outcomes is estimated using the following empirical specification:

$$\Delta y_{d,2000-2010} = \alpha + \beta_1 \Delta OutputImportTariff_d + \beta_2 \Delta InputImportTariff_d + \beta_3 \Delta ExportTariff_d + \gamma X_{d,2}$$

where:

- $\Delta y_{d,2000-2010}$ denotes the change in the district-level labor market outcome between 2000 and 2010.
- $\Delta OutputImportTariff_d$ is the district-level output import tariff exposure.
- $\Delta InputImportTariff_d$ is the district-level input import tariff exposure.
- $\Delta ExportTariff_d$ is the district-level export tariff exposure.
- $X_{d,2000}$ is a vector of pre-liberalization district characteristics, including industrial composition, education, demographic variables, and island fixed effects.
- ε_d is the error term.

To address the potential endogeneity of tariff reductions, Delesalle et al. (2025) instrument district-level tariff changes using the initial average tariff level in 2000. The identification strategy exploits the design of the ACFTA, under which sectors with higher pre-liberalization tariffs experienced systematically larger tariff reductions between 2000 and 2010. Consequently, the initial tariff level provides a strong predictor of subsequent tariff changes while remaining plausibly exogenous to later labor market outcomes after conditioning on pre-treatment characteristics.

Unlike Kis-Katos and Sparrow (2015), who rely on a first-difference OLS specification, Delesalle et al. (2025) estimate the causal effects using a Bartik IV strategy, thereby strengthening causal identification by mitigating potential bias arising from the non-random allocation of tariff reductions across industries.

3. Metodologi

3.1 Data

3.2 Metode

4. Hasil

5. Kesimpulan

6. Referensi

6.1 References

Add your bibliography entries to `ref.bib` and cite them using standard Quarto citation syntax. Here is an example citation on diamond open access (**fuchs2013diamond?**). According to (**fuchs2013diamond?**), diamond open access model is amazing.

6.2 Footnotes

Use footnotes sparingly. They should be used for additional information that is not essential to the main text.[1]

6.3 Tables

Use standard markdown tables or Quarto's table syntax. Tabel 1 shows an example.

Tabel 1. Example table

Parameter	Notation	Remarks
name	-	engine common identifier
manufacture	-	name of the manufacturer
bpr	λ	bypass ratio

[1]This is an example footnote.

Parameter	Notation	Remarks
pr	-	pressure ratio
thrust	T_0	maximum static thrust

6.4 Figures

Use standard Quarto figure syntax with `![caption](path){#fig-id width=X%}`. Place images in a `fig/` folder.

6.5 Equations

Use standard LaTeX equation syntax. Persamaan 1 shows an example equation.

$$\rho \frac{\mathrm{D}\mathbf{u}}{\mathrm{D}t} = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \rho \, \mathbf{g} \tag{1}$$

7. Sections

Organize your paper using standard markdown headings.

Some standard sections are:

- Introduction
- Methods
- Results
- Discussion
- Conclusion

Acknowledgement

Include your acknowledgements in this section.

Funding Statement

When applicable, please specify the funding information for this research.

Reproducibility Statement

Mandatory section!

7.1 Information on how to reproduce this research, including access to: 1) source code related to the research, 2) source code for the figures, 3) source code/data for the tables when applicable.

8. Pendahuluan

9. Studi Literatur

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10. Metodologi

10.1 Data

10.2 Metode

11. Analisis